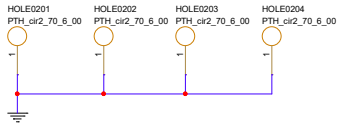
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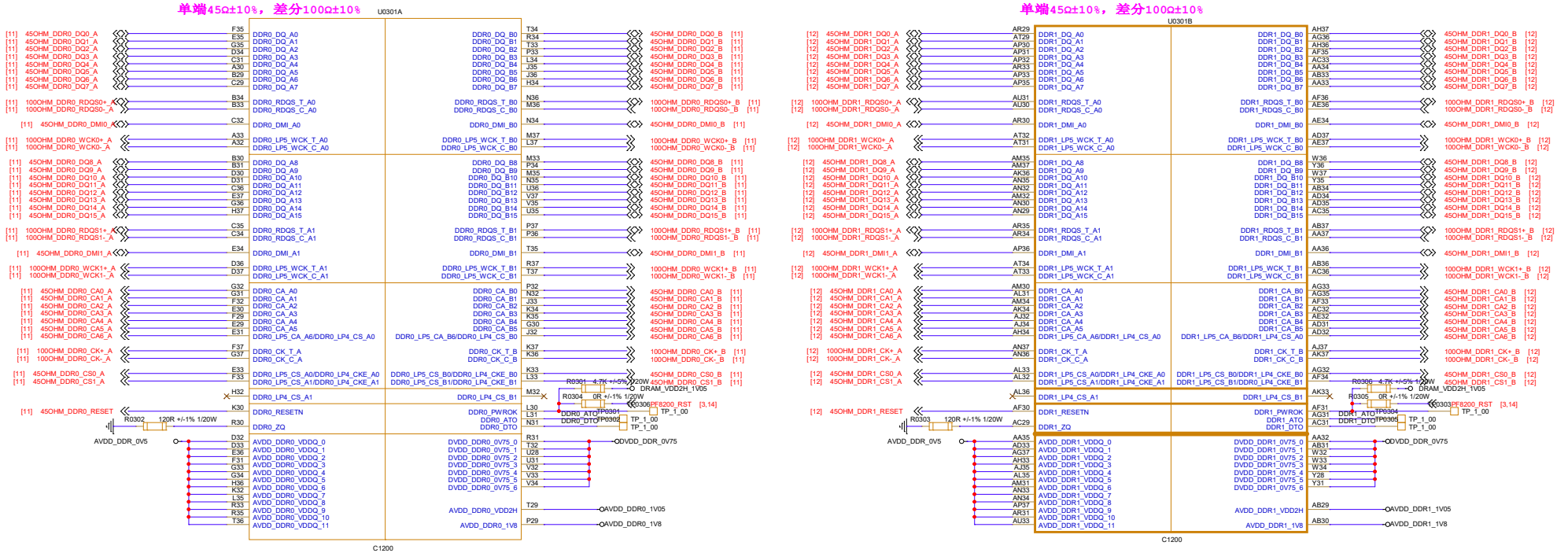
Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
SIZE: A2	DATE: Friday, January 16, 2026	SHEET: 1	OF: 18

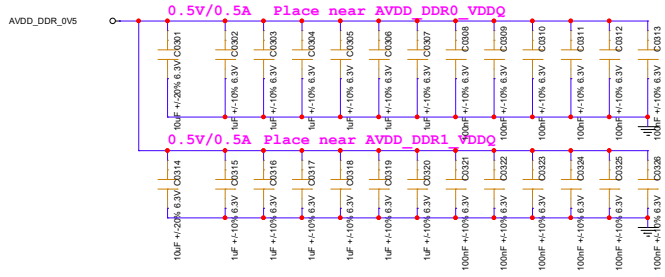
Version	Date	Author	Change List	Approved
V1.1	2025-11	Alan.Ren	First edition for BST C1200 CORE V1.1	Terence.Tang

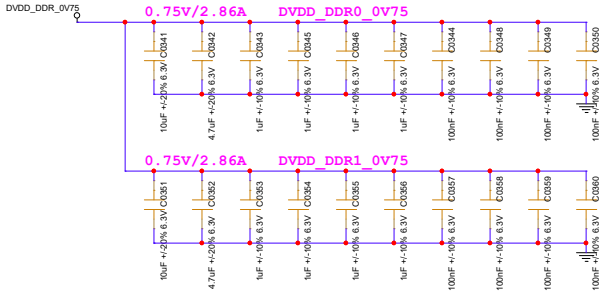


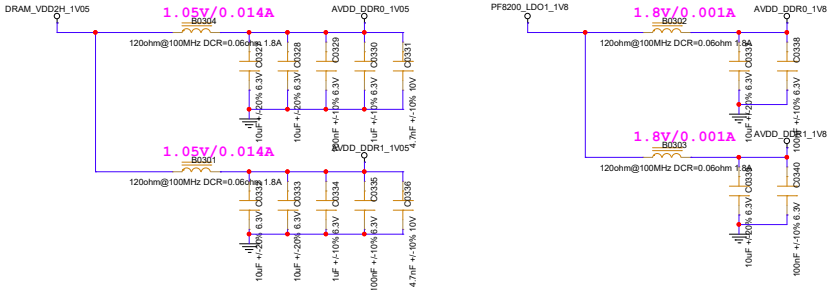
Quectel Wireless Solutions

PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren			CHECKED BY: Terence
SIZE: A2	DATE: Friday, January 16, 2026	SHEET: 2	OF: 18

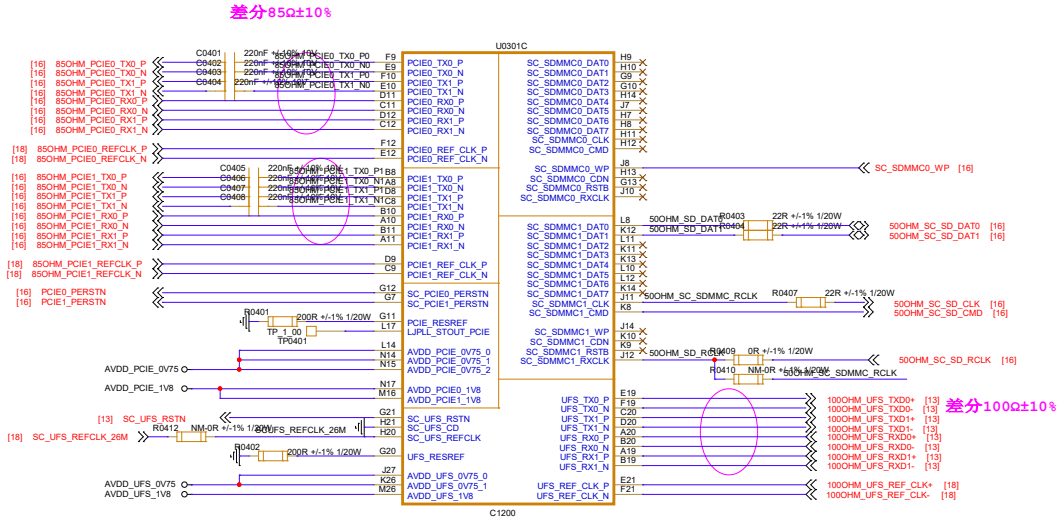








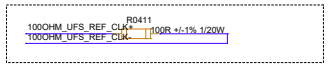
Quectel Wireless Solutions			
PROJECT	QSC801XP-AP		VER v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 3	OF 18



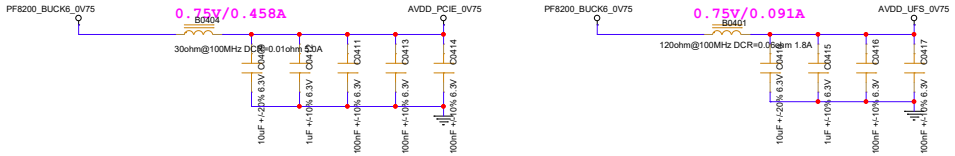
Config Table

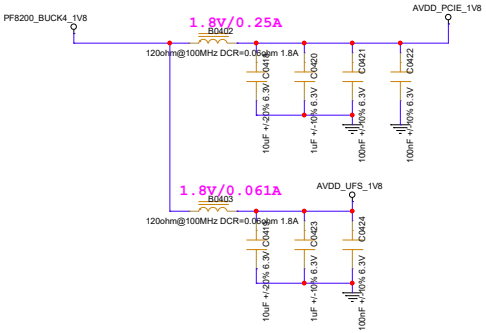
	GPIO (Default)	SD Card
R0410	Remove	Install
R0409	Install	Remove

R0410 & R0409 need to be placede close to the SOC.

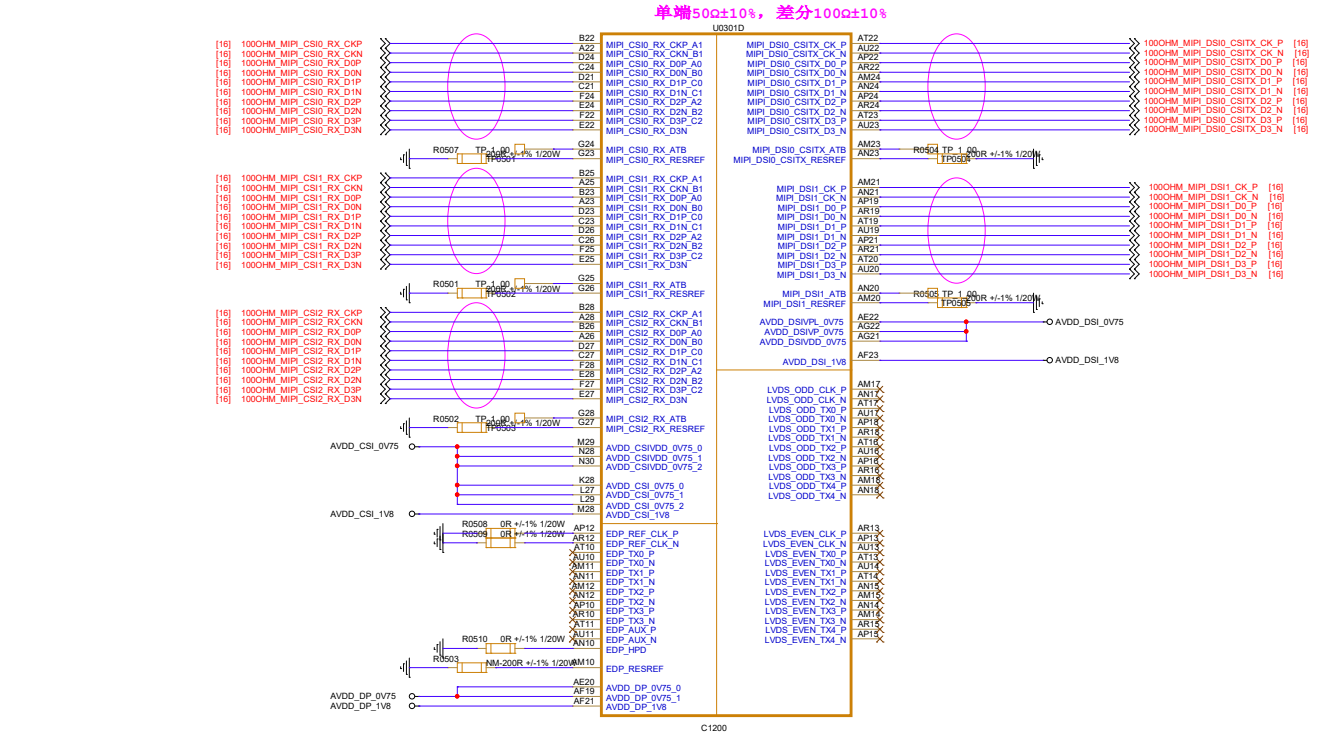


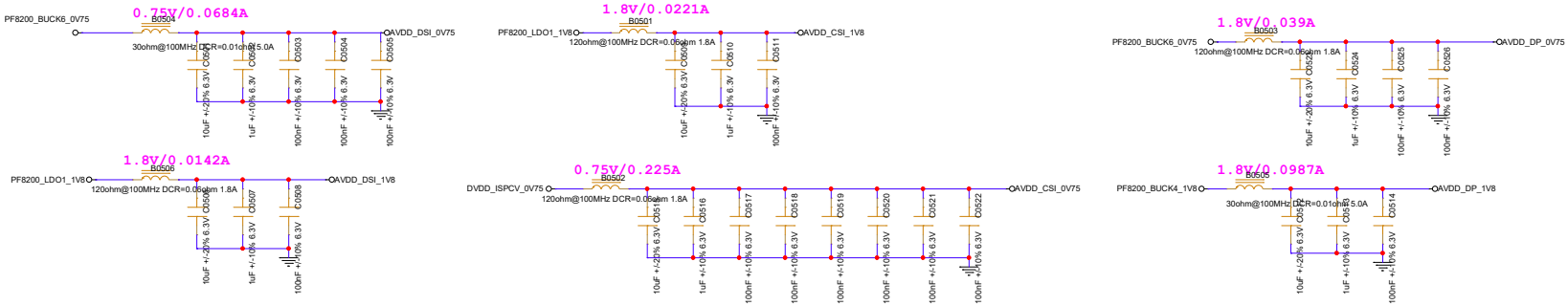
R0411 need to be place close to the UFS_REF_CLK PAD.





Quectel Wireless Solutions			
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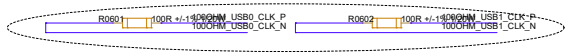


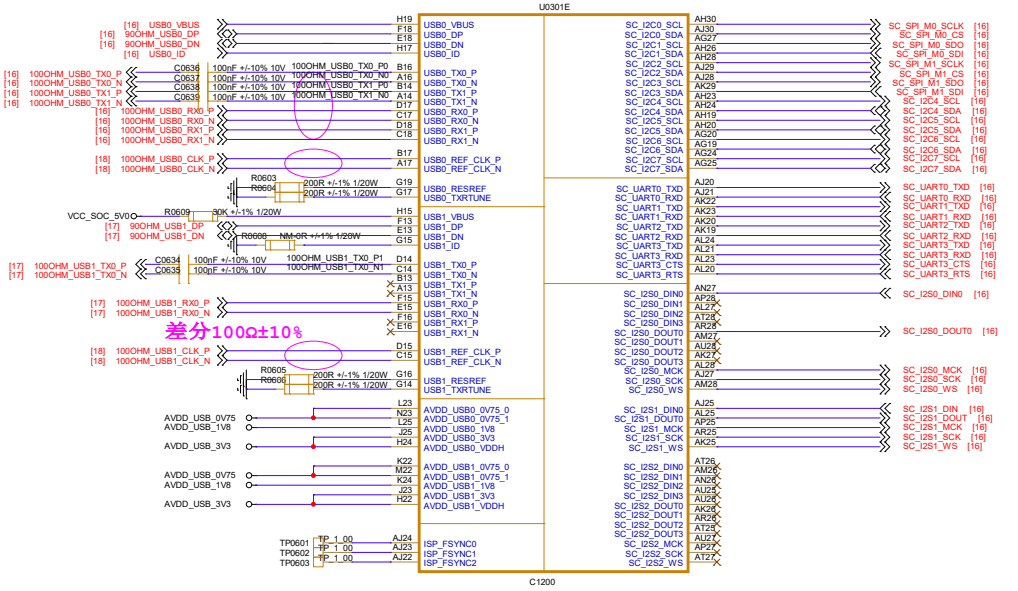


Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
SIZE: A2	DATE: Friday, January 16, 2026	SHEET: 5	OF: 18

R0601&R0602 need to be place close to the USB_REF_CLK PAD.

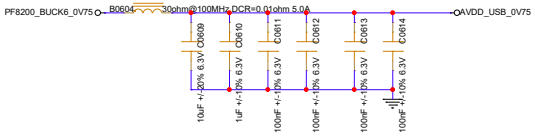
差分100Ω±10%



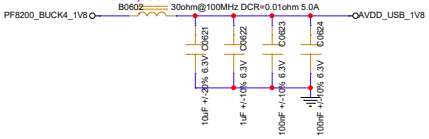


差分100Ω±10%

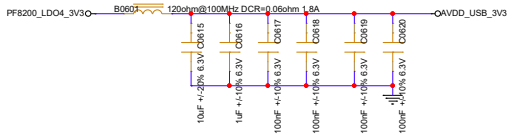
0.75V/0.196A



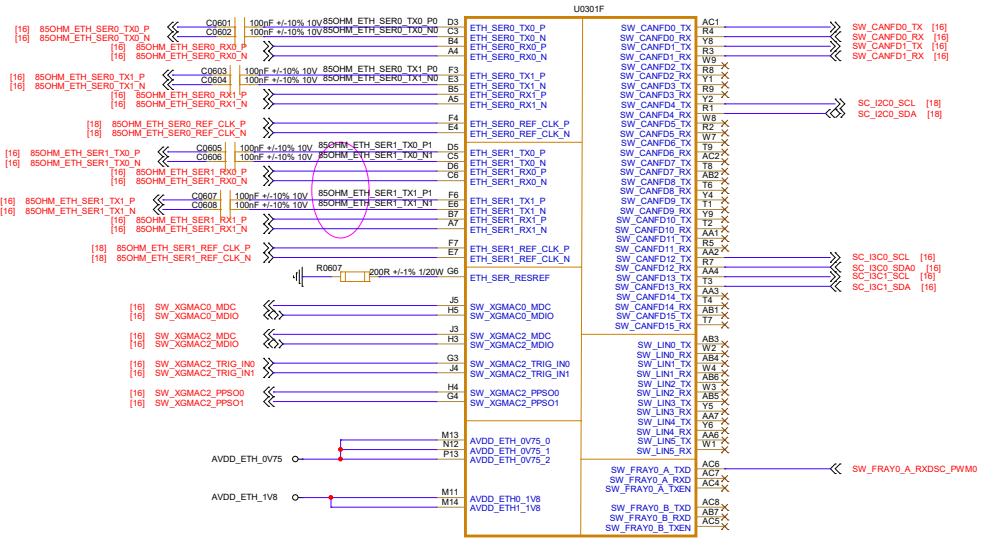
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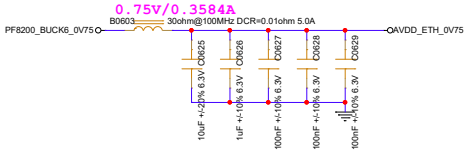
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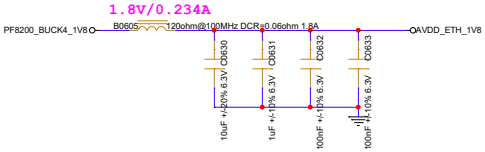
差分85Ω±10%



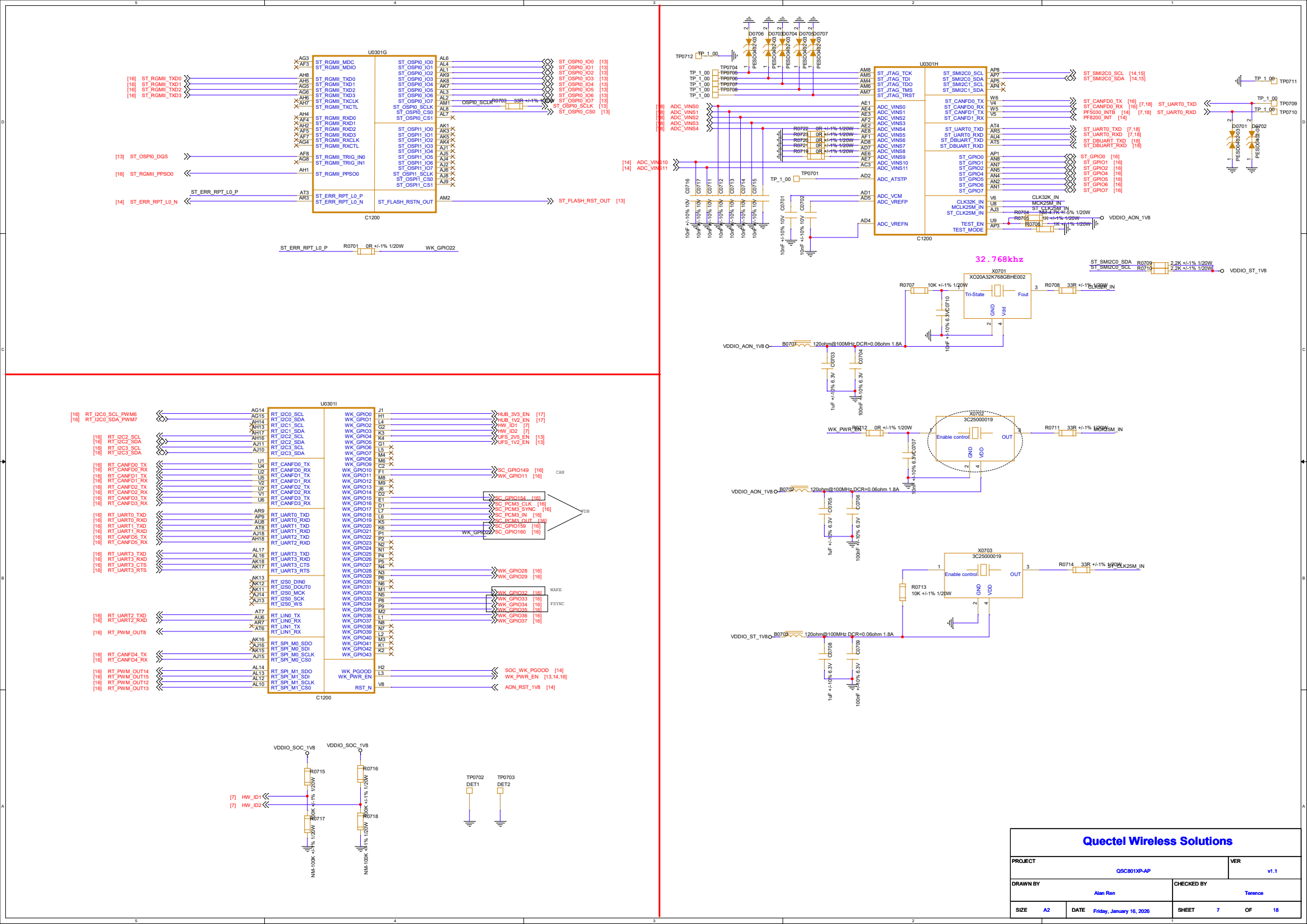
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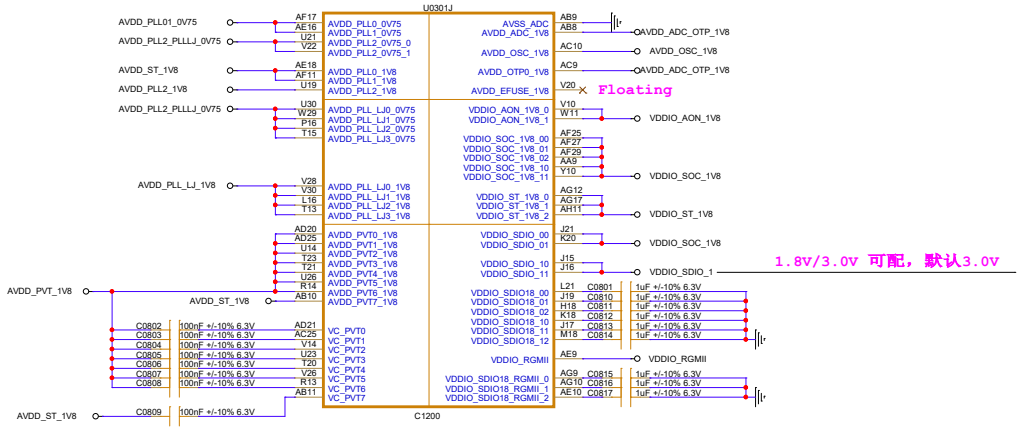
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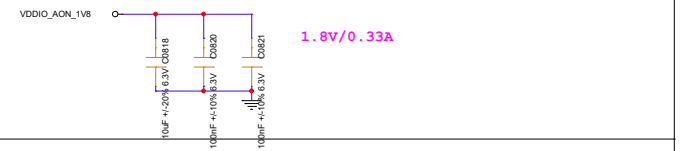


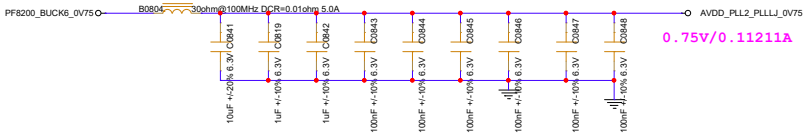
Quectel Wireless Solutions			
PROJECT	QSC801XP-AP	VER	v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 6	OF 18

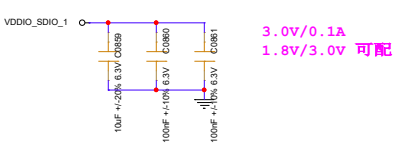

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C1200
U0301H
C1200
U0301I
C1200
32.768khz
X0701
XO20A32K768GBHE002
X0702
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X0703
3C25000019
Quectel Wireless Solutions
PROJECT
QSC801XP-AP
VER
v1.1
DRAWN BY
Alan Ren
CHECKED BY
Terence
SIZE
A2
DATE
Friday, January 16, 2026
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7
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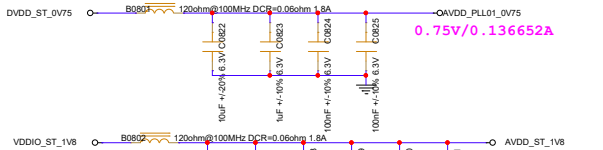
SOC PWR #1

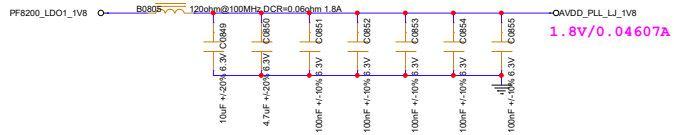


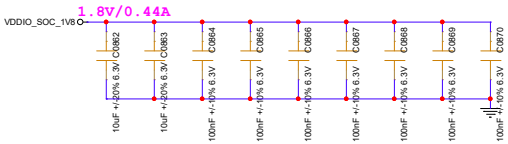


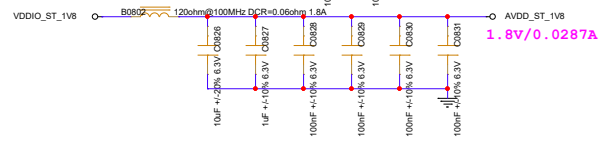


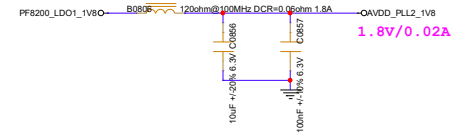


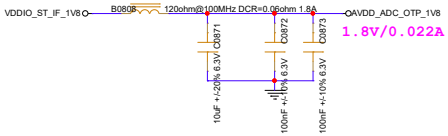


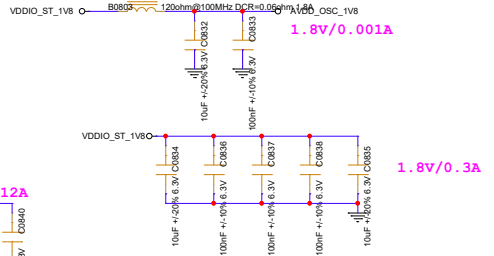


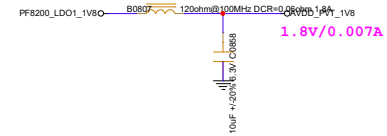


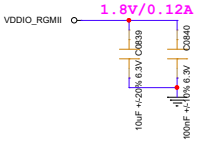






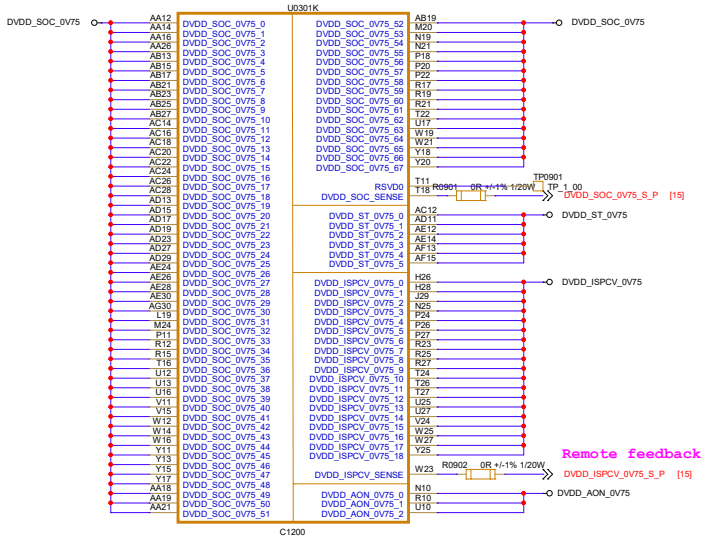


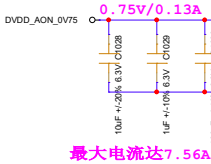


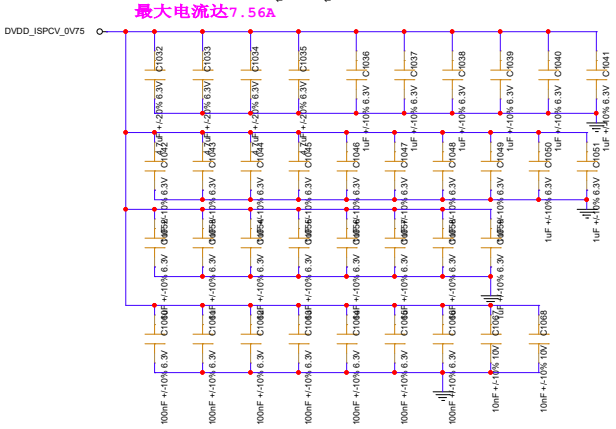


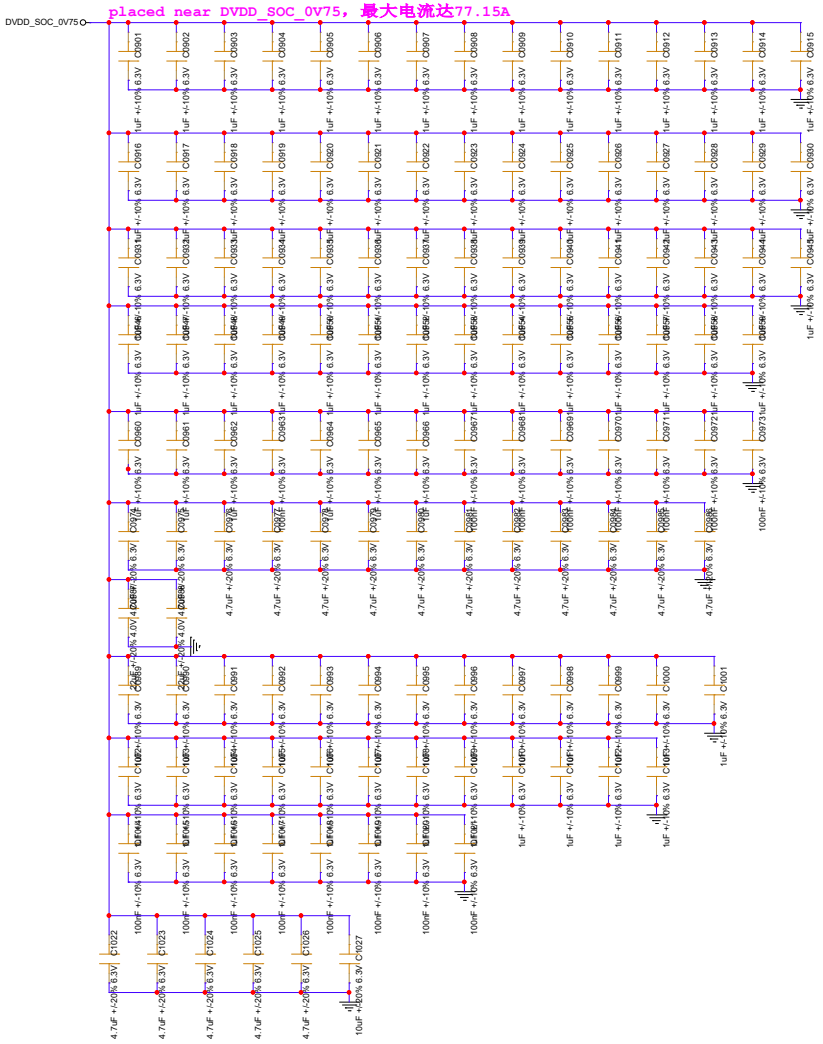
Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
SIZE: A2	DATE: Friday, January 16, 2026	SHEET: 8	OF: 18

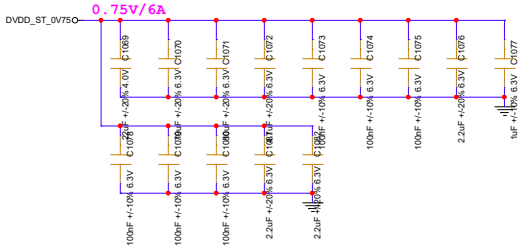
SOC PWR #2



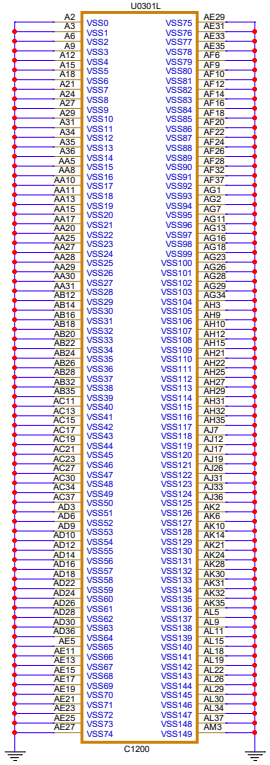


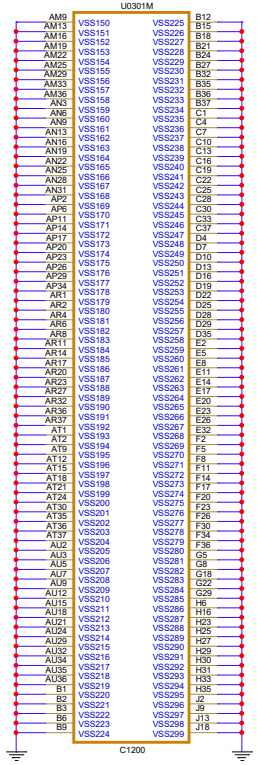


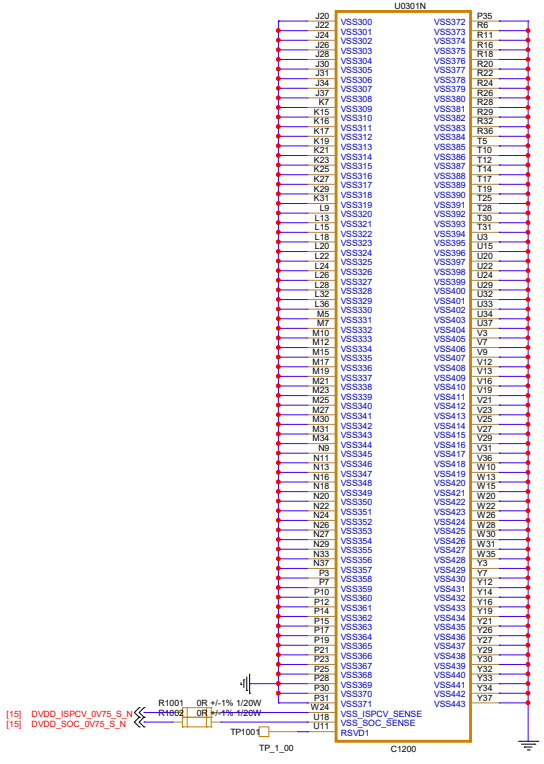




Quectel Wireless Solutions			
PROJECT	QSC801XP-AP		VER v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 9	OF 18

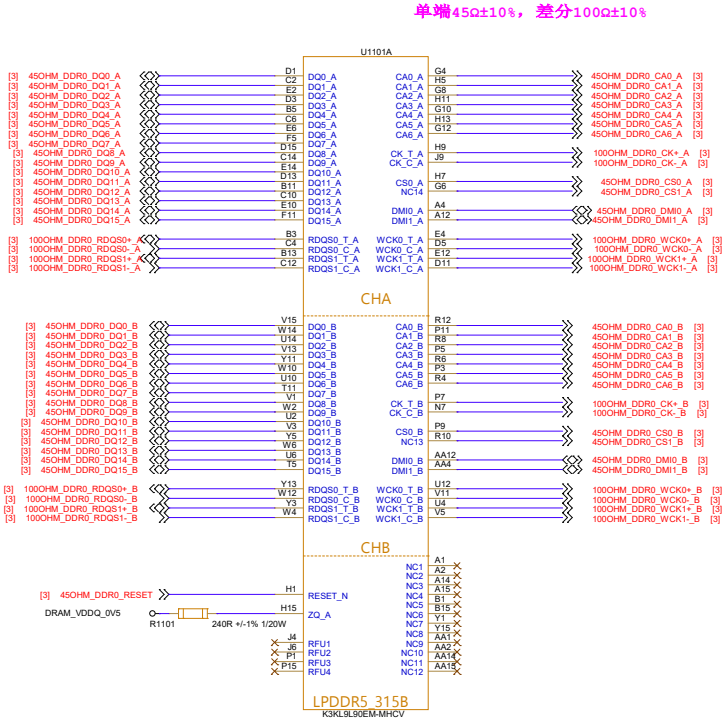


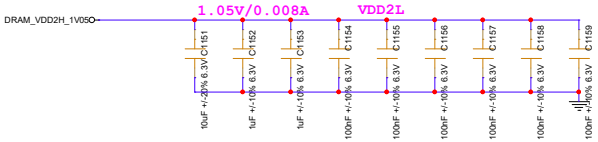


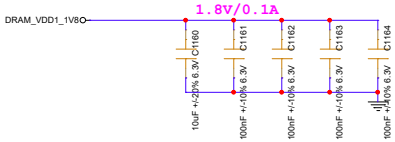


Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
SIZE: A2	DATE: Friday, January 16, 2026	SHEET: 10	OF: 18

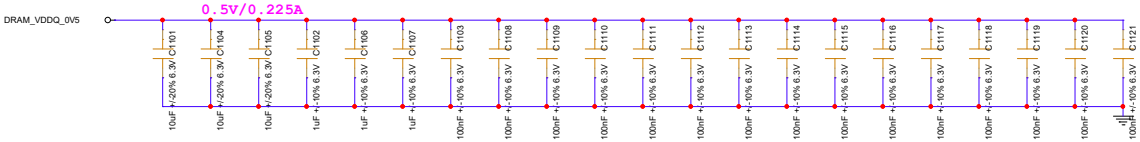
单端45Ω±10%，差分100Ω±10%

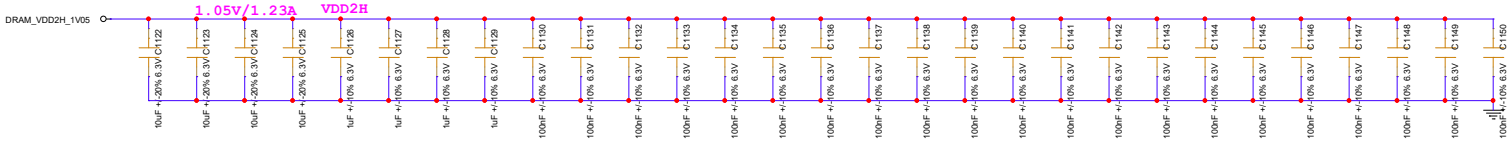


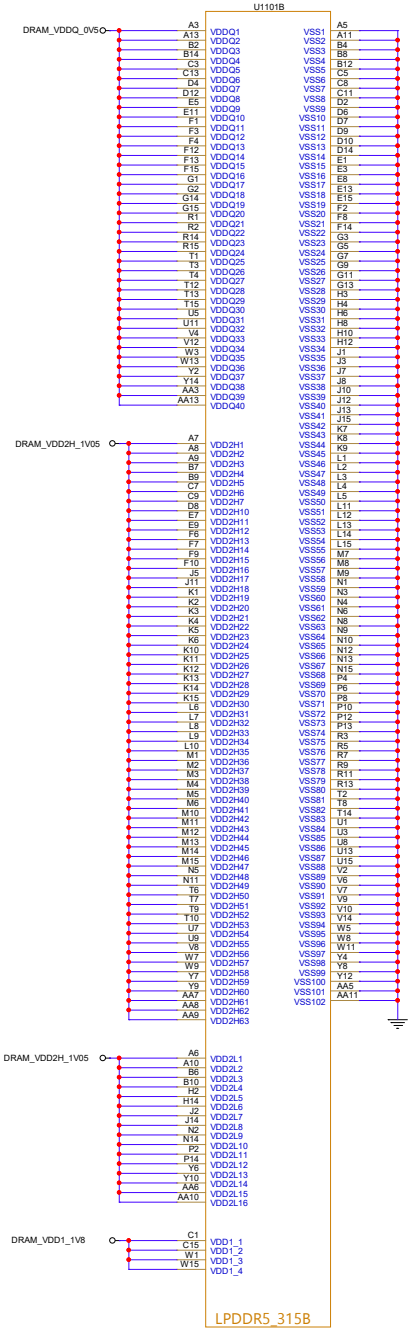




NOTE:
DRAM各路最大电流按K3KL9L90EM-MHCV评估，若使用的是其它型号的颗粒，需重新评估

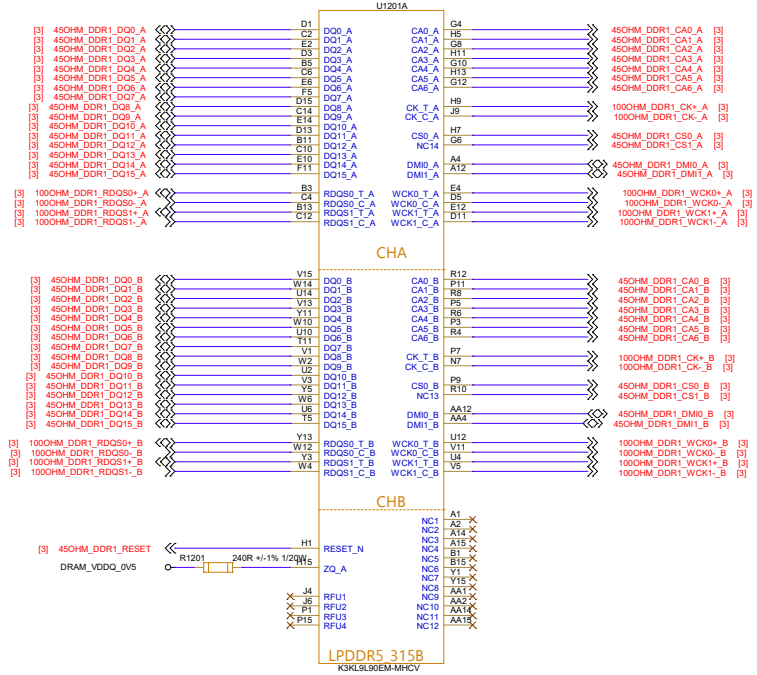


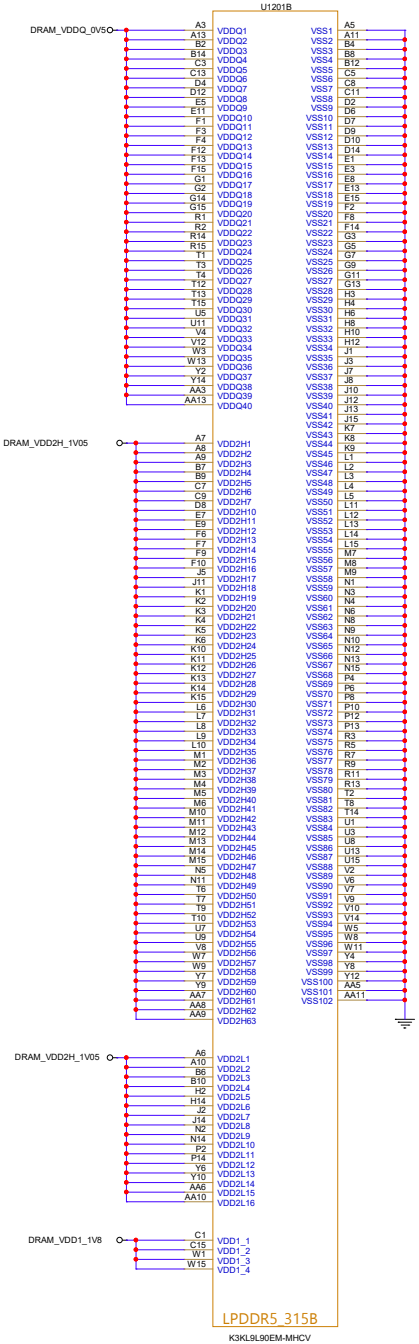




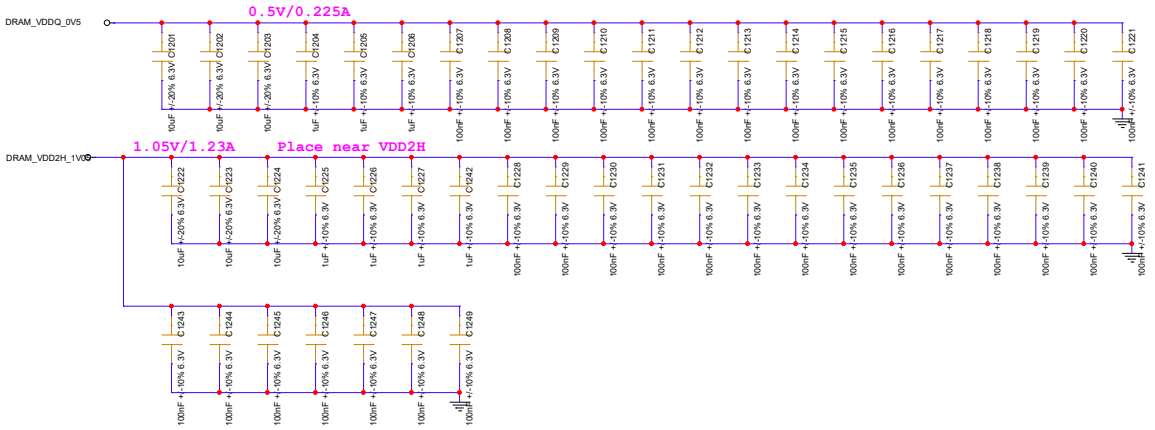
Quectel Wireless Solutions			
PROJECT	QSC801XP-AP		VER v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 11	OF 18

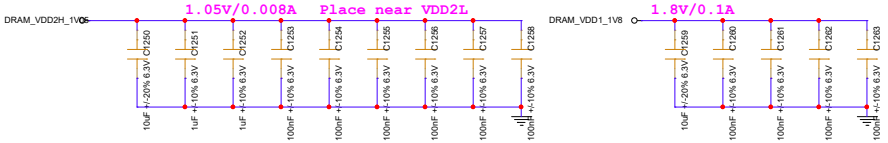
单端45Ω±10%，差分100Ω±10%



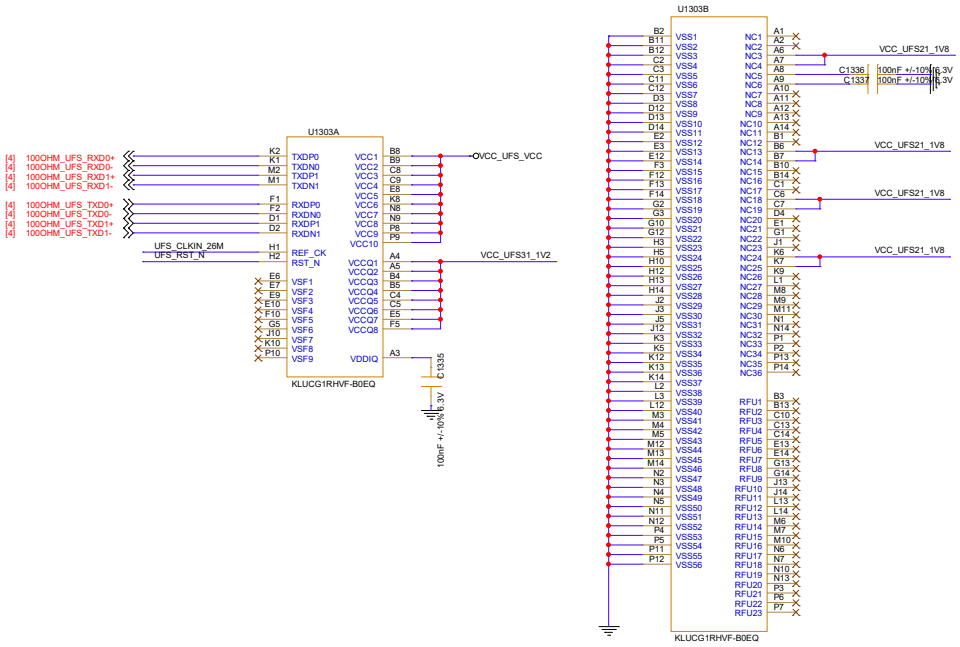


NOTE:
DRAM各路最大电流按K3KL9L90EM-MHCV评估，若使用的是其它型号的颗粒，需重新评估

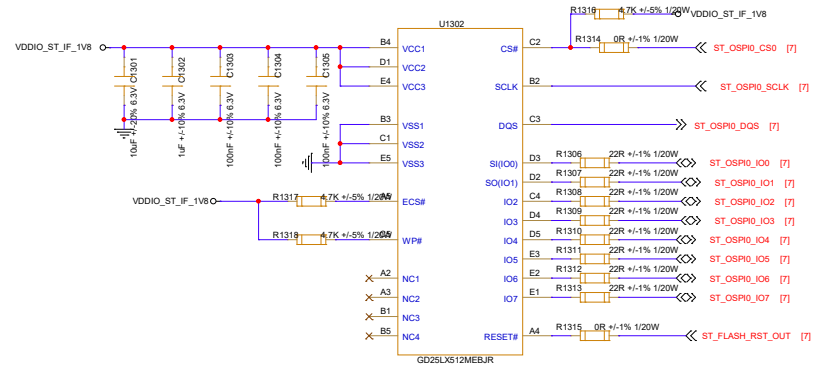


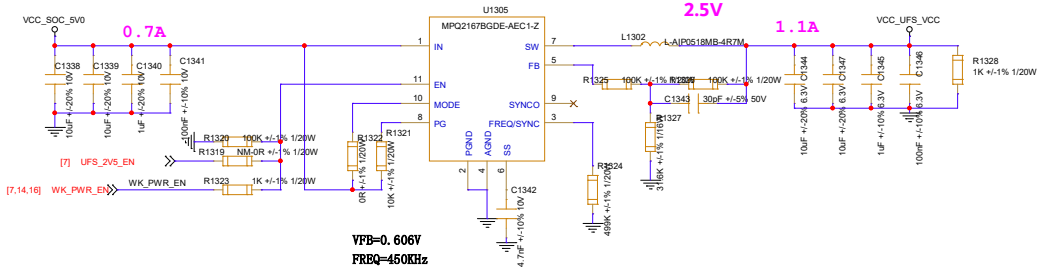


Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
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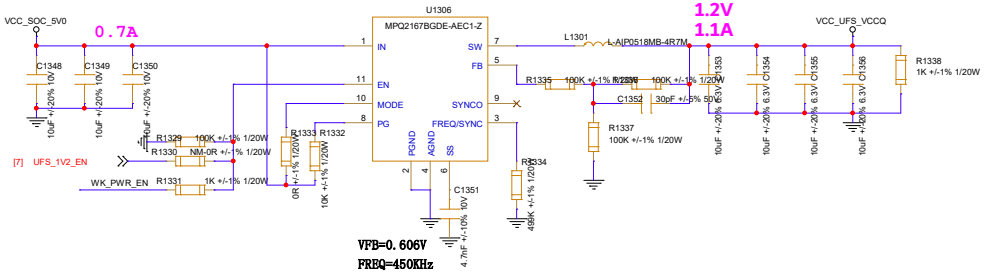


OSPI NOR FLASH



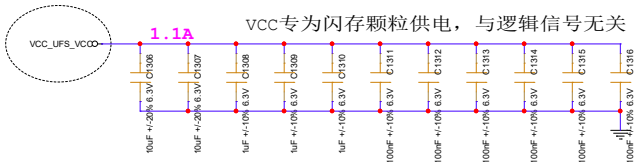


VCC_SOC_5V0-->VCC_UFS_2V5



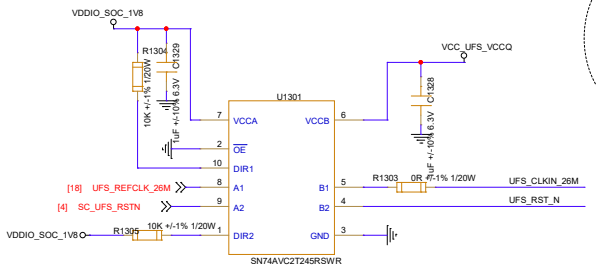
VCC_SOC_5V0-->VCC_UFS_1V2

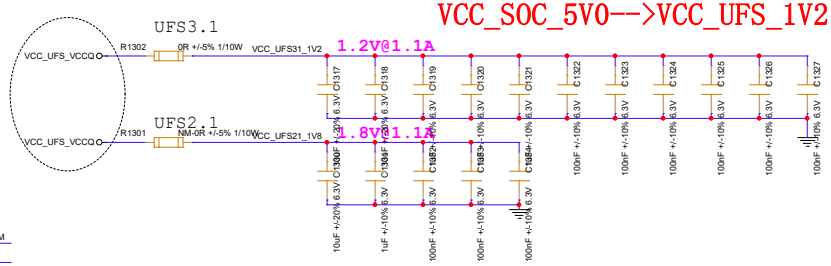
UFS3.1 NAND FLASH



VCC专为闪存颗粒供电，与逻辑信号无关

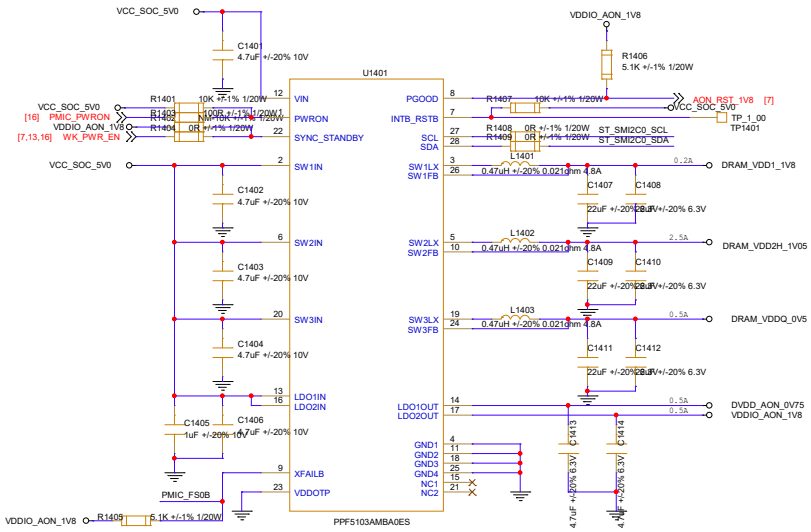
选用UFS2.1时需要：
调整VCC_UFS_VCC由2.5V改为3.3V。

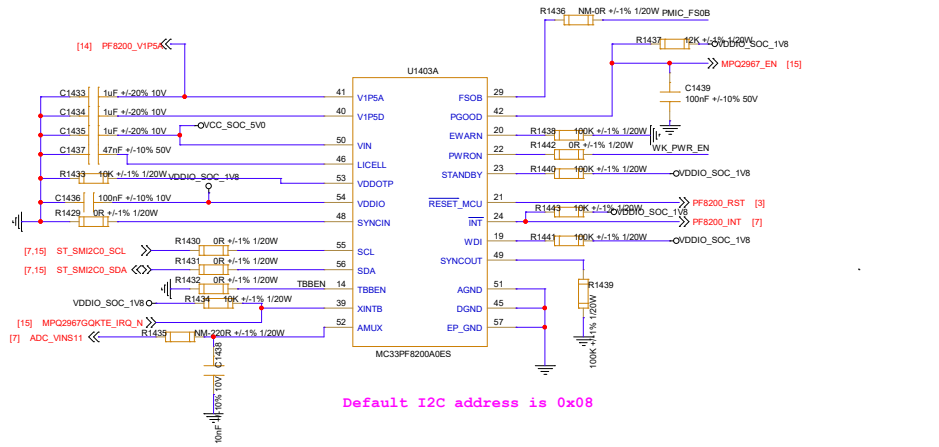


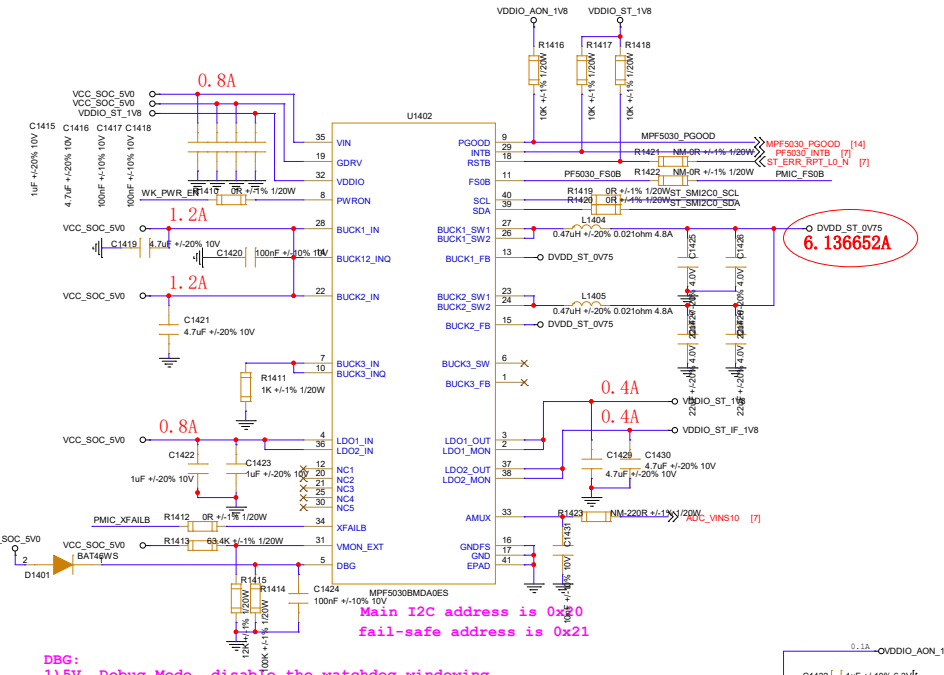


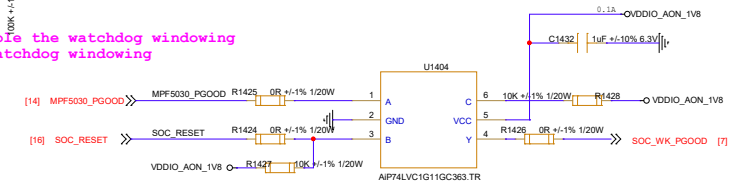
选用UFS2.1时需要：
贴装R1301，空贴R1302，然后将VCC_UFS_VCCQ由1.2V改为1.8V。

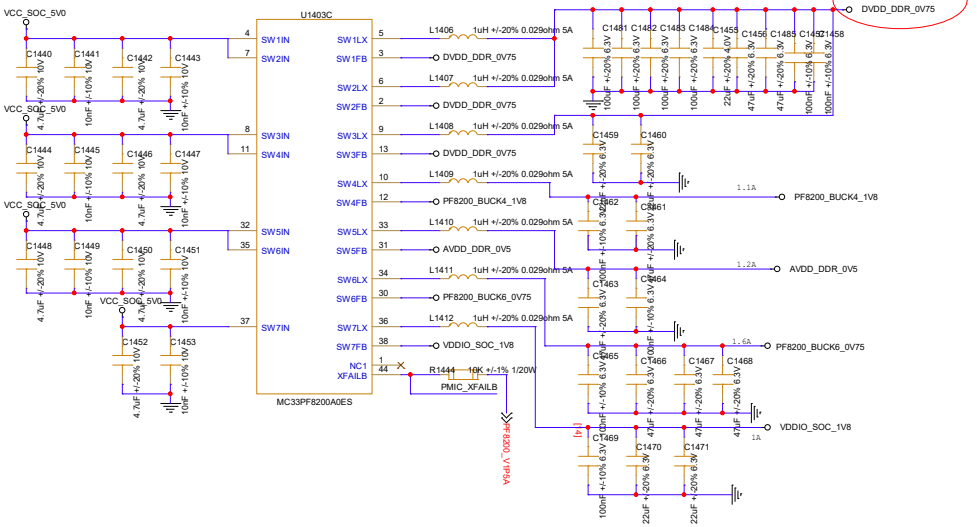
Quectel Wireless Solutions			
PROJECT	QSC801XP-AP	VER	v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 13	OF 18

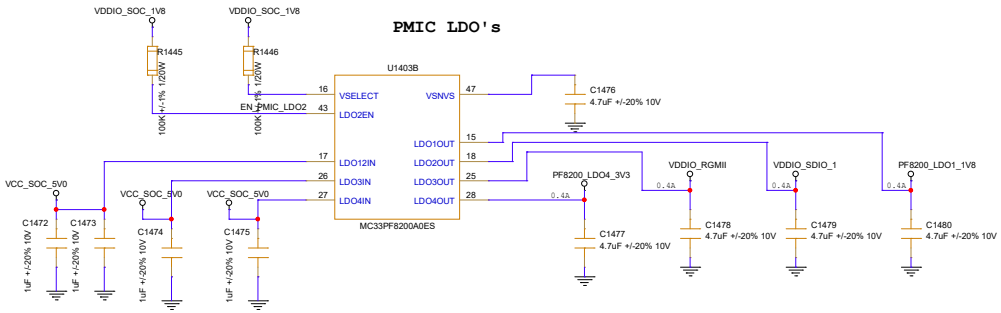




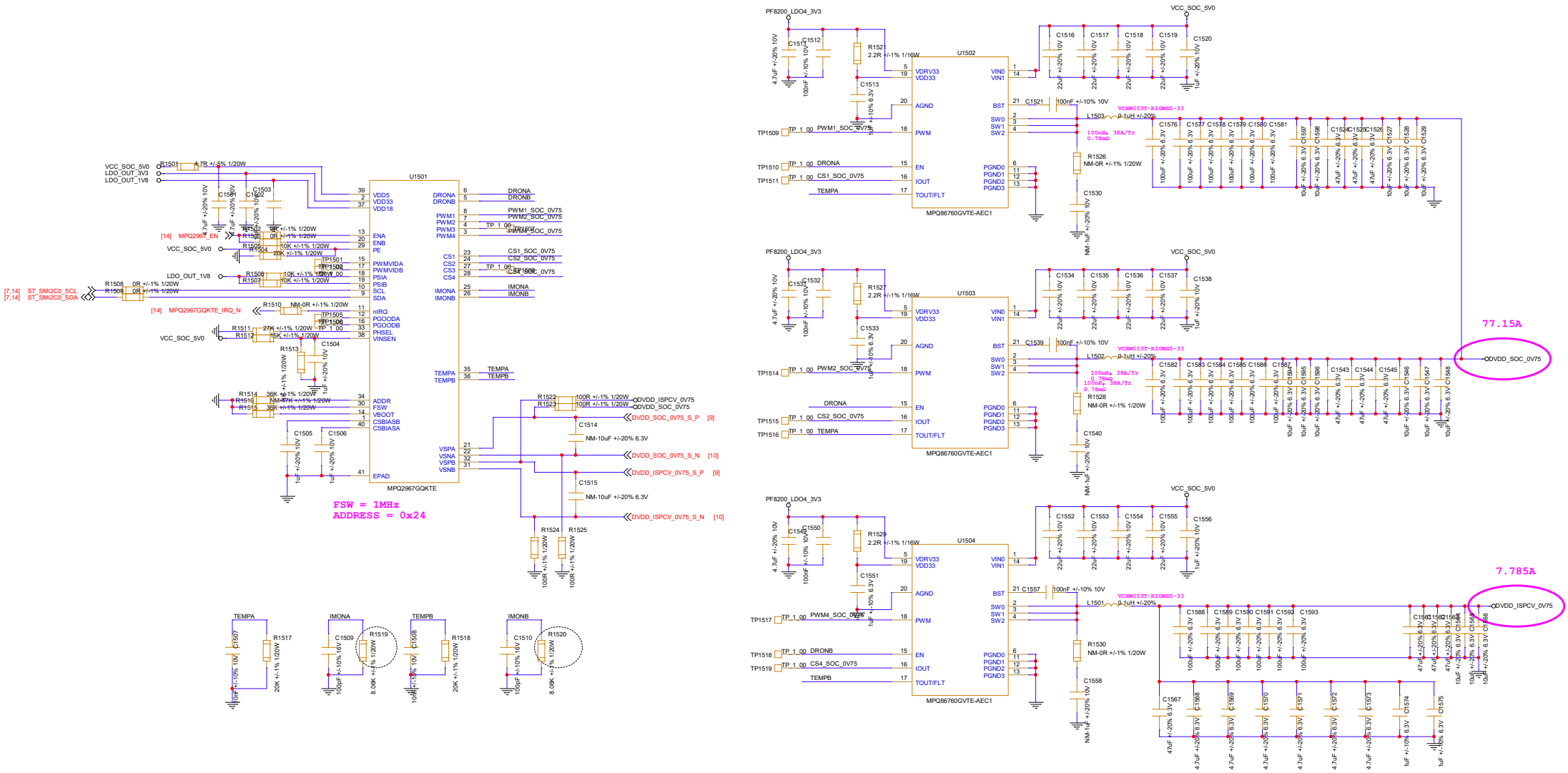




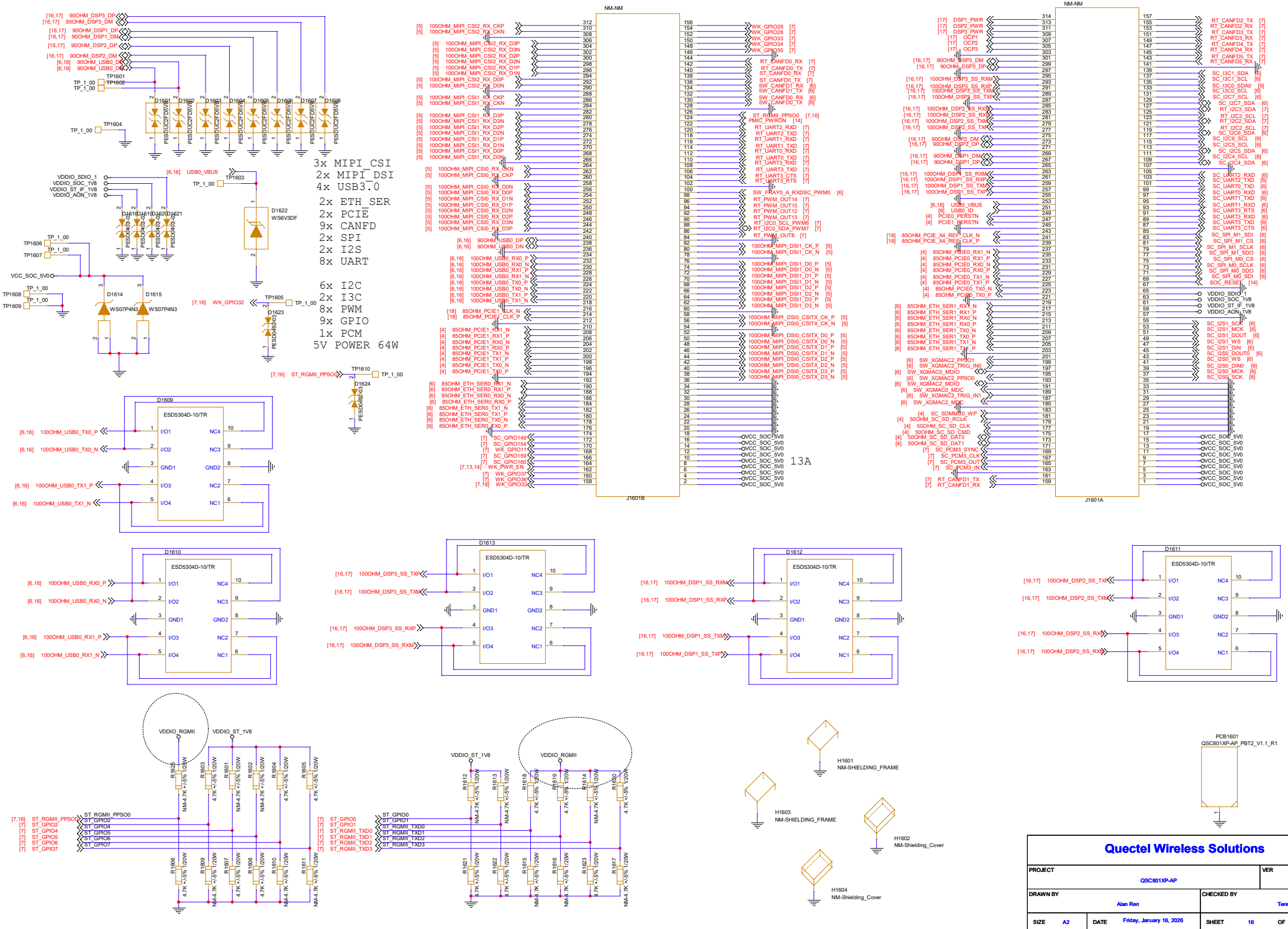


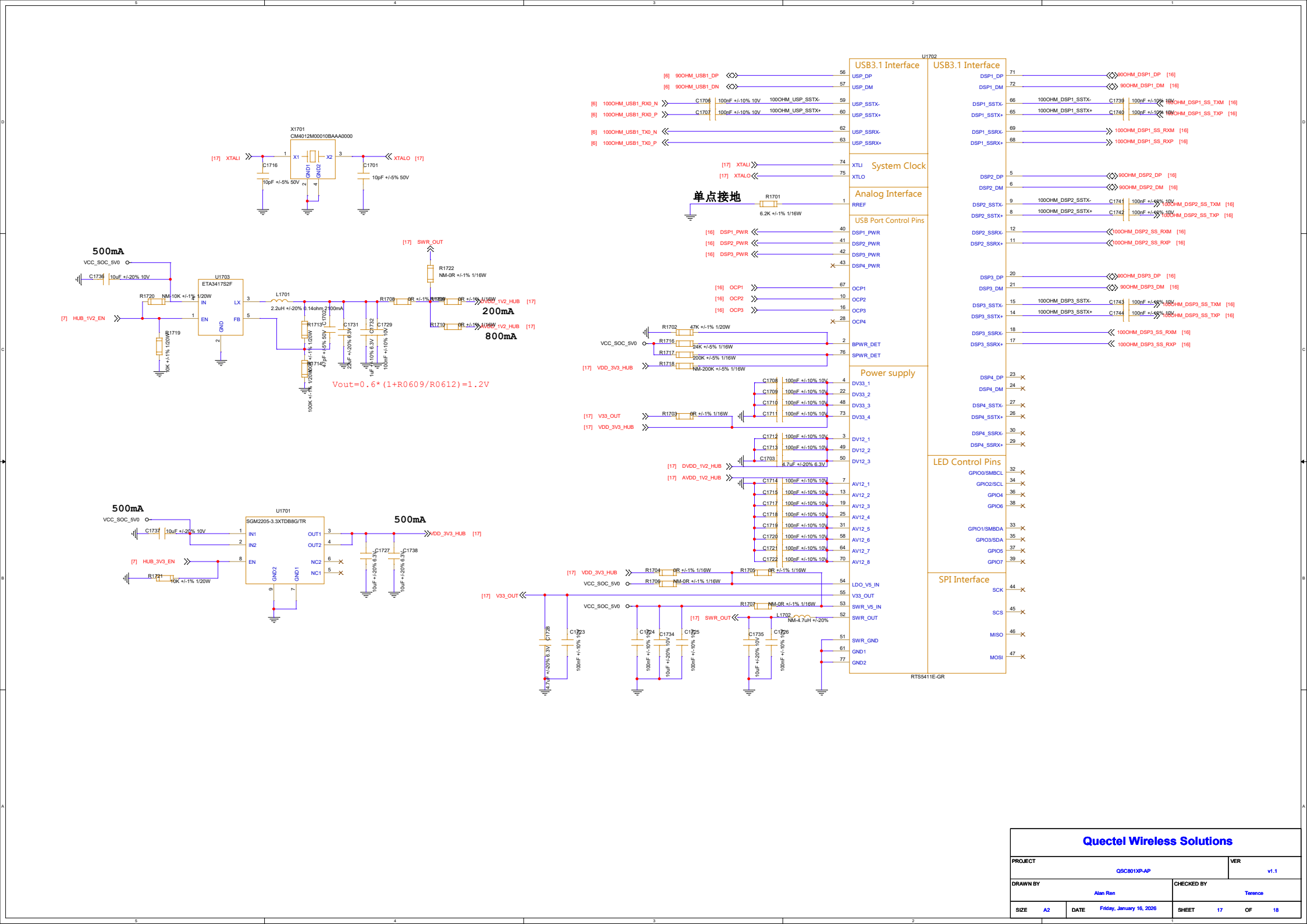


Quectel Wireless Solutions			
PROJECT	QSC801XP-AP	VER	v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 14	OF 18

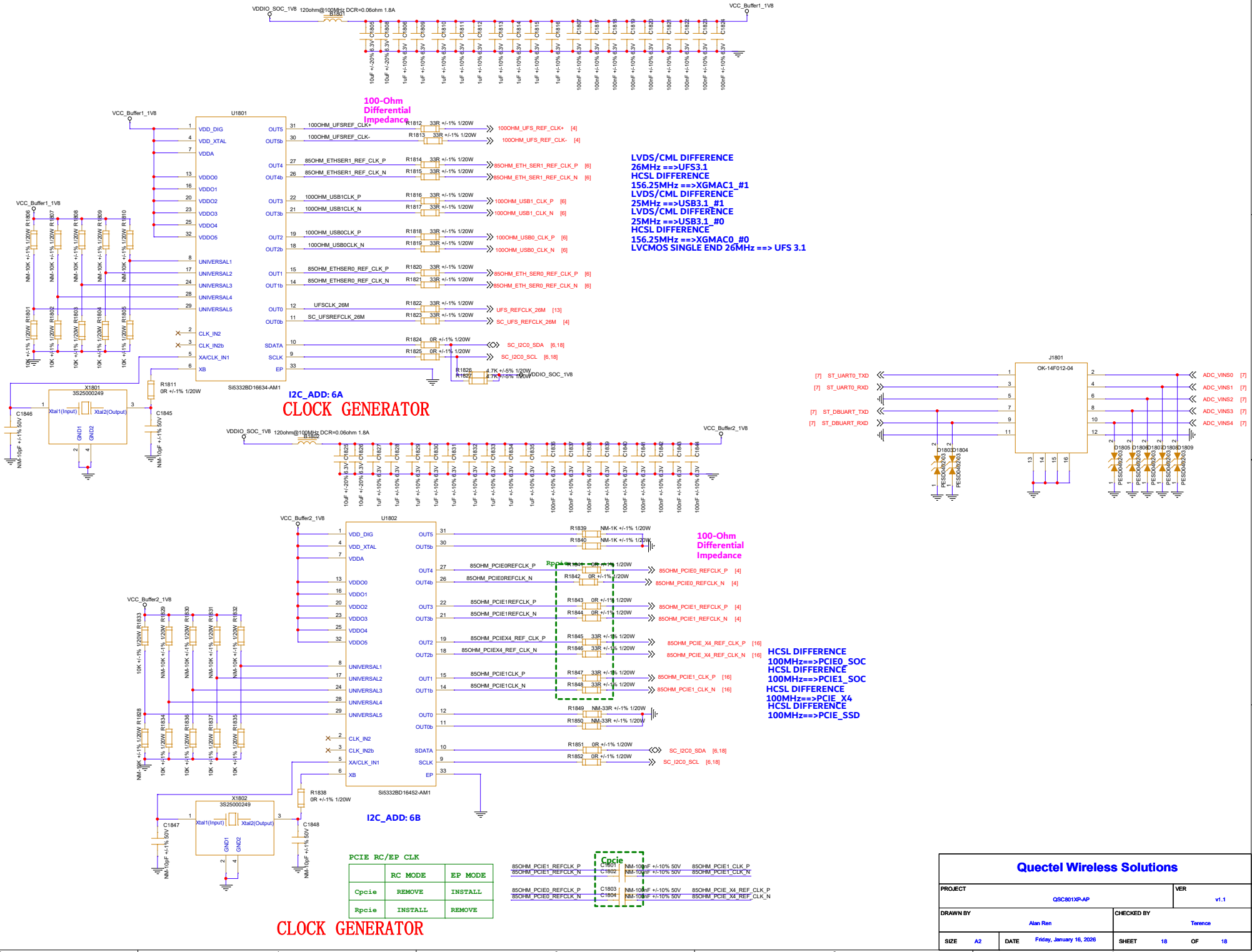


Quectel Wireless Solutions			
PROJECT: QSC801XP-AP			VER: v1.1
DRAWN BY: Alan Ren		CHECKED BY: Terence	
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3x MIPI_CSI
2x MIPI_DSI
4x USB3.0
2x ETH_SER
2x PCIE
9x CANFD
2x SPI
2x I2S
8x UART
6x I2C
2x I3C
8x PWM
9x GPIO
1x PCM
5V POWER 64W
13A
NM-NM
J1601B
J1601A
ESD5304D-10/TR
H1601
NM-SHIELDING_FRAME
H1603
NM-SHIELDING_FRAME
H1602
NM-Shielding_Cover
H1604
NM-Shielding_Cover
PCB1601
QSC801XP-AP_PBT2_V1.1_R1
Quectel Wireless Solutions
PROJECT
QSC801XP-AP
VER
v1.1
DRAWN BY
Alan Ren
CHECKED BY
Terence
SIZE
A2
DATE
Friday, January 16, 2026
SHEET
16
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18



Quectel Wireless Solutions			
PROJECT	QSC801XP-AP		VER v1.1
DRAWN BY	Alan Ren	CHECKED BY	Terence
SIZE A2	DATE Friday, January 16, 2026	SHEET 17	OF 18


VDDIO_SOC_1V8
120ohm@100MHz DCR=0.06ohm 1.8A
R1801
VCC_Buffer1_1V8
100-Ohm Differential Impedance
U1801
VDD_DIG
VDD_XTAL
VDDA
VDDO0
VDDO1
VDDO2
VDDO3
VDDO4
VDDO5
UNIVERSAL1
UNIVERSAL2
UNIVERSAL3
UNIVERSAL4
UNIVERSAL5
CLK_IN2
CLK_IN2b
XA/CLK_IN1
XB
OUT5
OUT5b
OUT4
OUT4b
OUT3
OUT3b
OUT2
OUT2b
OUT1
OUT1b
OUT0
OUT0b
SDATA
SCLK
EP
100OHM_UFSREF_CLK+
100OHM_UFSREF_CLK-
85OHM_ETHSER1_REF_CLK_P
85OHM_ETHSER1_REF_CLK_N
100OHM_USB1CLK_P
100OHM_USB1CLK_N
100OHM_USB0CLK_P
100OHM_USB0CLK_N
85OHM_ETHSER0_REF_CLK_P
85OHM_ETHSER0_REF_CLK_N
UFSCLK_26M
SC_UFSREFCLK_26M
SC_I2C0_SDA
SC_I2C0_SCL
Si5332BD16634-AM1
X1801
3S25000249
Xtal1(Input)
Xtal2(Output)
GND1
GND2
I2C_ADD: 6A
CLOCK GENERATOR
LVDS/CML DIFFERENCE
26MHz ==>UFS3.1
HCSL DIFFERENCE
156.25MHz ==>XGMAC1_#1
LVDS/CML DIFFERENCE
25MHz ==>USB3.1_#1
LVDS/CML DIFFERENCE
25MHz ==>USB3.1_#0
HCSL DIFFERENCE
156.25MHz ==>XGMAC0_#0
LVCMOS SINGLE END 26MHz ==> UFS 3.1
J1801
OK-14F012-04
ST_UART0_TXD
ST_UART0_RXD
ST_DBUART_TXD
ST_DBUART_RXD
ADC_VINS0
ADC_VINS1
ADC_VINS2
ADC_VINS3
ADC_VINS4
PESD0402/03
PESD0402/05
VCC_Buffer2_1V8
R1802
U1802
Si5332BD16452-AM1
X1802
I2C_ADD: 6B
HCSL DIFFERENCE
100MHz==>PCIE0_SOC
HCSL DIFFERENCE
100MHz==>PCIE1_SOC
HCSL DIFFERENCE
100MHz==>PCIE_X4
HCSL DIFFERENCE
100MHz==>PCIE_SSD
PCIE RC/EP CLK
RC MODE
EP MODE
Cpcie
REMOVE
INSTALL
INSTALL
REMOVE
CLOCK GENERATOR
Quectel Wireless Solutions
PROJECT
QSC801XP-AP
VER
v1.1
DRAWN BY
Alan Ren
CHECKED BY
Terence
SIZE
A2
DATE
Friday, January 16, 2026
SHEET
18
OF
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